CH-8

$$Z_0 = \frac{d}{w}\sqrt{\frac{\mu}{\varepsilon}} \qquad \varepsilon = \varepsilon_r.\varepsilon_0$$

a) $\varepsilon_r \times 2$. $Z_0 \to \sqrt{2}$ küçülür. $\frac{d}{w}$ oranı $\sqrt{2}$ büyümeli

b) $\frac{w}{\sqrt{2}}$ olmalı

c) $w = 2w$ olmalı

d) Yayılma hızı $u_p = \frac{1}{\sqrt{\mu\varepsilon}}$

8.2) → $R = \frac{2}{w}\sqrt{\frac{\pi f \mu_0}{\sigma_c}} = \frac{2}{0{,}02}\sqrt{\frac{\pi.500\times10^6 \times 4\pi\times10^{-7}}{1{,}6\times10^7}}$

$\mu_0 = 4\pi\times10^{-7}$ H/m

$R = 1{,}11\ \Omega/m$

→ $L = \mu\frac{d}{w} = (4\pi\times10^{-7}).\frac{2{,}5\times10^{-3}}{20\times10^{-3}} = 0{,}157\ \mu H/m$

→ $G = \sigma\frac{w}{d} = 10^{-3}.\frac{0{,}02}{2{,}5\times10^{-3}} = 0{,}008\ S/m$

→ $C = \varepsilon\frac{w}{d} = \varepsilon_r.\varepsilon_0\frac{w}{d} = 3.\ 8{,}85\times10^{-12}.\frac{0{,}02}{2{,}5\times10^{-3}} = 0{,}212\times10^{-9}\ f/m$

$\varepsilon_0 = 8{,}85\times10^{-12}$

b) $\frac{|E_z|}{|E_y|} = \sqrt{\frac{\overset{2\pi f}{w}.\varepsilon}{\sigma_c}} = \sqrt{\frac{2\times\pi\times500\times10^6\times8{,}85\times10^{-12}\times3}{1{,}6\times10^7\ (S/m)}} = 4{,}167\times10^{-5}$

c) önce wL ile R'yi
wC " G'yi
kontrol ederiz.

$wL = \underbrace{2\pi\times500\times10^6\times0{,}157\times10^{-6}}_{493{,}5\ \gg\ ✓} \gg 1{,}11$

$wC = \underbrace{2\pi\times500\times10^6\times0{,}212\times10^{-9}}_{0{,}667\ ✓} \gg 0{,}008$

Ana formüller;

$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$

ama eğer $\omega L \gg R$
$\omega C \gg G$ ise şu pratik formülleri kullan

$\gamma = \alpha + j\beta$

→ $\alpha = \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}$

→ $\beta = w\sqrt{LC}$

→ $Z_0 = \sqrt{\frac{L}{C}}$

$\gamma = 0{,}129 + j18{,}14\ m^{-1}$

$Z_0 = 27{,}21 + j013\ \Omega$

8.5) $Z_0 = 50 + j0\ \Omega$ 100 MHz

$\alpha = 0{,}01$ dB/m

$\beta = 0{,}8\pi$ rad/m R, L, C, G ?

$Z_0 = 50 + j0$ yani tamamen reel

Yani $\frac{R}{L} = \frac{G}{C}$) distortionless line / bozulmasız hat $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$

pratik formüller

$\alpha = R\sqrt{\frac{C}{L}}$

$\beta = w\sqrt{LC}$

$Z_0 = \sqrt{\frac{L}{C}}$

1) formüllerde α Np/m birimli olmalı

1 Np = 8,686 dB

$\alpha = \frac{0{,}01}{8{,}686} = 0{,}00115$ Np/m

2) $\alpha = \frac{R}{Z_0}$ $0{,}00115 = \frac{R}{50}$ $R = 0{,}0575\ \Omega$

→ $L = \frac{\beta Z_0}{\underset{w}{2\pi f}} = \frac{0{,}8\pi\times50}{2\times\pi\times10^8} = 0{,}2\times10^{-6}$ H/m

→ $C = \frac{L}{Z_0^2} = \frac{0{,}20\times10^{-6}}{50^2} = 80\times10^{-12}$

→ $G = ?$ $\frac{R}{L} = \frac{G}{C}$ $G = \frac{R.C}{L} = \frac{0{,}0575 . 80\times10^{-12}}{0{,}2\times10^{-6}}$

$G = 23\times10^{-6}$ S/m

8.7) $V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$

$I(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z}$

a) Exponential formu:

$V(z) = \frac{1}{2}(V_i + I_i Z_0)e^{-\gamma z} + \frac{1}{2}(V_i - I_i Z_0)e^{\gamma z}$ → $V_0^+ = \frac{1}{2}(V_i + I_i.Z_0)$ $V_0^- = \frac{1}{2}(V_i - I_i.Z_0)$

$I(z) = \frac{1}{2Z_0}(V_i + I_i Z_0)e^{-\gamma z} - \frac{1}{2Z_0}(V_i - I_i Z_0)e^{\gamma z}$

b) Hyperbolic form:

$V(z) = V_i\cosh(\gamma z) - I_i Z_0 \sinh(\gamma z)$

$I(z) = I_i\cosh(\gamma z) - \frac{V_i}{Z_0}\sinh(\gamma z)$

8.8) Write governing voltage and current transmission line equations.

$\frac{-dV}{dz} = (R + \cancel{j\omega L})I$

$\frac{-dI}{dz} = (G + \cancel{j\omega C})V$

Bu soruda DC old. için

$\frac{-dV}{dz} = R.I$ $\frac{d^2V}{dz^2} = RGV$

$\frac{-dI}{dz} = G.V$ $\frac{d^2I}{dz^2} = RGI$

b) $V(z) = V_0^+ e^{-\overset{\alpha}{\gamma} z} + V_0^- e^{\overset{\alpha}{\gamma} z}$ $\alpha = \sqrt{R.G}$

$I(z) = I_0^+ e^{-\overset{\alpha}{\gamma} z} + I_0^- e^{\overset{\alpha}{\gamma} z}$

$\gamma = \alpha + \cancel{j\beta}$

$\frac{V_0^+}{I_0^+} = \frac{V_0^-}{I_0^-} = R_0 = \sqrt{\frac{R}{G}}$

c) Infinite line: geri dönen dalga yok V_0^-

$V(z) = V_0^+ e^{-\gamma z} = V_0^+ e^{-\alpha z}$

$I(z) = \frac{V_0^+}{R_0} e^{-\alpha z}$

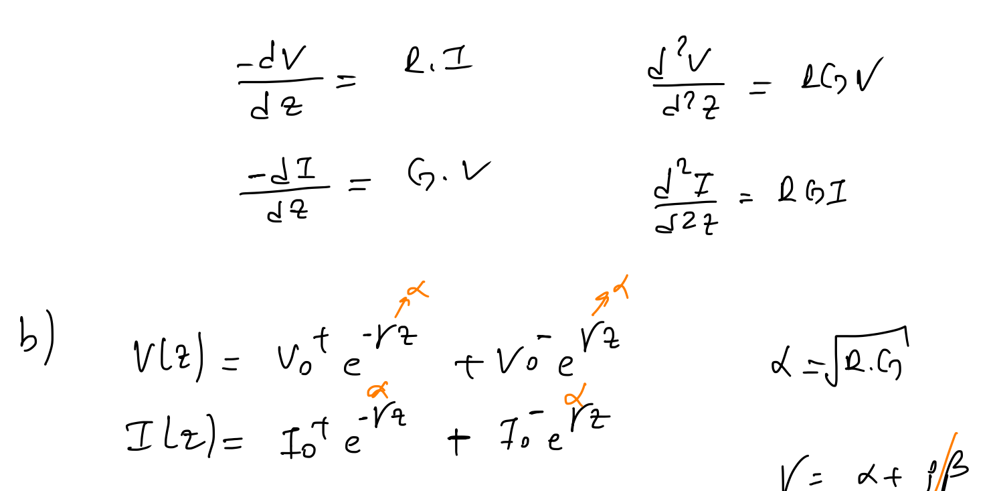
$V(z) = \frac{V_g.R_0}{R_g + R_0} e^{-\alpha z}$ $I(z) = \frac{V_g}{R_g + R_0} e^{-\alpha z}$

d) Finite line, lenght ℓ, load R_L)

$Z_{in} = R_0\frac{R_L + R_0\tanh(\alpha\ell)}{R_0 + R_L\tanh(\alpha\ell)}$

kaynak akımı: $I_i = \frac{V_g}{R_g + Z_{in}}$ $V_i = I_i . Z_{in}$

$$\boxed{\begin{aligned} V(z) &= V_i.\cosh(\alpha z) - I_i R_0 \sinh(\alpha z)\\ I(z) &= I_i.\cosh(\alpha z) - \frac{V_i}{R_0}\sinh(\alpha z)\end{aligned}}$$



P.8-9 A generator with an open-circuit voltage $v_g(t) = 10\sin 8000\pi t$ (V) and internal impedance $Z_g = 40 + j30$ (Ω) is connected to a 50-(Ω) distortionless line. The line has a resistance of 0.5 (Ω/m), and its lossy dielectric medium has a loss tangent of 0.18%. The line is 50 (m) long and is terminated in a matched load. Find (a) the instantaneous expressions for the voltage and current at an arbitrary location on the line, (b) the instantaneous expressions for the voltage and current at the load, and (c) the average power transmitted to the load.

R_0

R

P.8-9 Distortionless line: $R_0 = \sqrt{\frac{L}{C}} = 50\ (\Omega)$, $R = 0.5\ (\Omega/m)$,

$\tan\left(\frac{\sigma}{\omega\epsilon}\right) = \tan\left(\frac{G}{\omega C}\right) = 0.0018.$

$\longrightarrow \frac{G}{\omega C} \cong 0.0018;\ \frac{G}{C} = 8000\pi\times0.0018 = 45.2 = \frac{R}{L}.$

$L = \frac{R}{G/C} = 0.011\ (H/m),\quad C = \frac{L}{R_0^2} = 4.42\ (\mu F/m).$

$\alpha = \frac{R}{R_0} = 0.010\ (Np/m),\quad \beta = \omega\sqrt{LC} = 5.55\ (rad/m).$

a) $V(z) = \frac{V_{g0}R_0}{R_0 + R_g}e^{-\alpha z}e^{-j\beta z} = \frac{50}{9+j3}e^{-0.01z}e^{-j5.55z};\quad I(z) = \frac{V(z)}{50}.$

$\therefore V(z,t) = 5.27\,e^{-0.01z}\sin(8000\pi t - 5.55z - 0.322)$ (V),

$I(z,t) = 0.105\,e^{-0.01z}\sin(8000\pi t - 5.55z - 0.322)$ (A).

b) At $z = 50$ (m): $V(50,t) = 3.20\sin(8000\pi t - 0.432\pi)$ (V),

$I(50,t) = 0.064\sin(8000\pi t - 0.432\pi)$ (A).

c) $(P_{av})_L = \frac{1}{2}\mathcal{Re}|V_L I_L^*| = \frac{1}{2}(3.20\times0.064) = 0.102$ (W).

$$\boxed{\begin{aligned} V_g(t) &= 10\sin(8000\pi t)\\ Z_g &= 40 + j30\ \Omega\end{aligned}}$$

$w = 8000\pi$

$R_0 = 50\ \Omega$

$\tan\delta = 0{,}18\times10^{-2}$

Distortionless line şartı:

$\frac{R}{L} = \frac{G}{C}$ $R_0 = \sqrt{\frac{L}{C}} = 50$

$\frac{G}{C} = w\tan\delta = 8000\pi . 0{,}0018 = 45{,}2$

$\frac{R}{L} = \frac{G}{C} = 45{,}2$

$\frac{R}{L} = 45{,}2$ $L = \frac{0{,}5}{45{,}2} = 0{,}011$ H/m

→ $R_0 = \sqrt{\frac{L}{C}} \Rightarrow 50 = \sqrt{\frac{0{,}011}{C}}$ $C = \frac{0{,}011}{2500} = 0{,}44\times10^{-5}$ f/m

α ve β'yı bulmamız lazım denklemde kullanmak için

$V(z) = \frac{V_{g0} . R_0}{R_0 + Z_g} e^{-\alpha z} . e^{-j\beta z}$

$\alpha = \frac{R}{R_0}$ $\beta = w\sqrt{LC}$

$\alpha = \frac{0{,}5}{50}$ $\beta = 8000\pi\sqrt{0{,}44\times10^{-5}\times0{,}011}$

$\alpha = 0{,}01$ $= 5{,}55$ rad/m

$V(z) = \frac{10 . 50}{50 + 40 + j30} e^{-0{,}01z} . e^{5{,}55j}$

$= \underbrace{\frac{500}{90 + j30}}_{5{,}27\angle-18{,}4^\circ} . e^{-0{,}01z - j5{,}55} = 5{,}27\angle-0{,}322\ rad . e^{-0{,}01z - j5{,}55}$

radyan $= \frac{\pi}{180} . 18{,}4 = -0{,}322$ rad (Bu sorularda radyan kullandığımız için radyana çevirdik)

$= 5{,}27e^{-0{,}01z} . e^{-0{,}322j - j5{,}55}$

Zamana çeviririz;

$V(z,t) = 5{,}27e^{-0{,}01z}\sin(8000t - 5{,}55z - 0{,}322)$

for current;

sadece reel kısmıyla işimiz

$I(z) = \frac{V(z)}{R_0} = \frac{V(z)}{50} = \frac{5{,}27}{50} = 0{,}105$

$i(z,t) = 0{,}105e^{-0{,}01z}\sin(8000\pi t - 5{,}55z - 0{,}322)$

b) Load'da $z = 50m$

$V(50,t) = 5{,}27e^{-0{,}01(50)}\sin(8000\pi t - 5{,}55(50) - 0{,}322)$

$V(50,t) = 3{,}2 . \sin(8000\pi t - 277{,}2)$

$I(50,t) = 0{,}105e^{-0{,}01(50)} . \sin(8000\pi t - 277{,}2)$

$I(50,t) = 0{,}064 . \sin(8000\pi t - 277{,}2)$

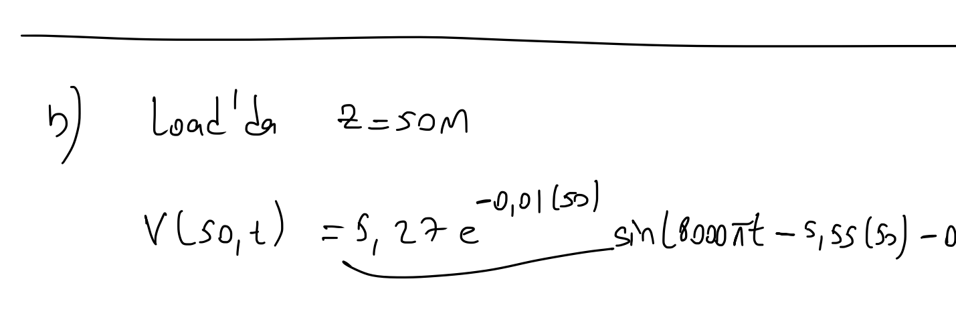
c) Average power?

Loaddaki max değerler $V = 3{,}2$

$I = 0{,}064$

$P_{av} = \frac{1}{2}V.I = \frac{1}{2} . 3{,}2\times0{,}064$

$P_{average} = 0{,}102\ W$



P.8-15 Consider a lossless transmission line.

a) Determine the line's characteristic resistance so that it will have a minimum possible standing-wave ratio for a load impedance 40 + j30 (Ω).

b) Find this minimum standing-wave ratio and the corresponding voltage reflection coefficient.

c) Find the location of the voltage minimum nearest to the load.

SWR

Z_L

Lossless line old. için $Z_0 = R_0$

P.8-15 For a lossless line, $Z_0 = R_0$.

$|\Gamma|^2 = \left|\frac{(R_L - R_0) + jX_L}{(R_L + R_0) + jX_L}\right|^2 = \frac{(R_L - R_0)^2 + X_L^2}{(R_L + R_0)^2 + X_L^2}.$

a) Set $\frac{\partial|\Gamma|^2}{\partial R_0} = 0. \longrightarrow R_0 = \sqrt{R_L^2 + X_L^2}.$

(A minimum S corresponds to a minimum $|\Gamma|$.)

For $Z_L = 40 + j30\ (\Omega)$, $R_0 = \sqrt{40^2 + 30^2} = 50\ (\Omega).$

b) $Min.\ |\Gamma| = \sqrt{\frac{R_0 - R_L}{R_0 + R_L}} = \sqrt{\frac{50-40}{50+40}} = \frac{1}{3}.$

$Min.\ S = \frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} = 2.$

lossless + Min SWR

a) $R_0 = ?$ $Z_L = \underset{40}{R_L} + j\underset{30}{X_L}$ Min SWR ⇒ $R_0 = \sqrt{R_L^2 + X_L^2}$

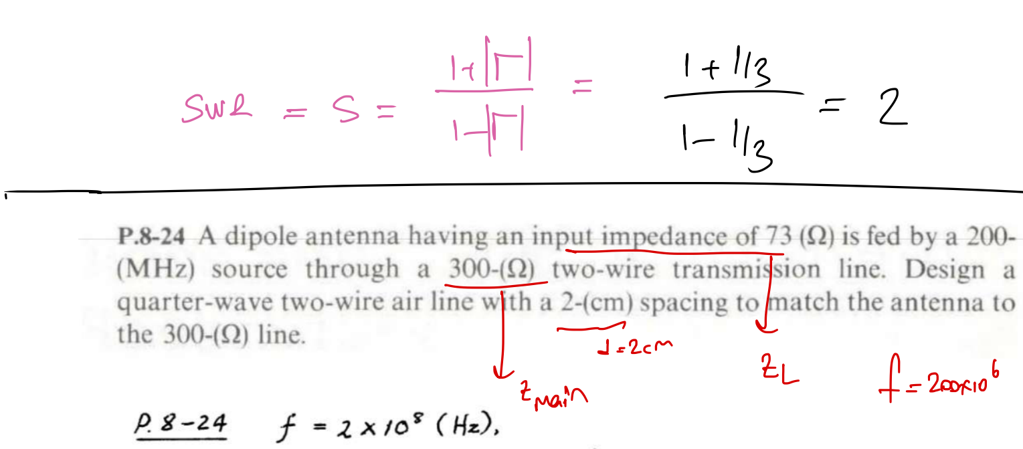
$R_0 = 50\ \Omega = Z_0$

b) Min SWR end reflection coefficient

$\frac{Z_L - Z_0}{Z_L + Z_0} = \frac{40 + j30 - 50}{40 + j30 + 50} = \frac{-10 + j30}{90 + j30}$

$|\Gamma| = \frac{\sqrt{10^2 + 30^2}}{\sqrt{90^2 + 30^2}} = \sqrt{\frac{1000}{8100 + 900}} = \sqrt{\frac{1000}{9000}} = \frac{1}{3}$

$SWR = S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 1/3}{1 - 1/3} = 2$



P.8-24 A dipole antenna having an input impedance of 73 (Ω) is fed by a 200-(MHz) source through a 300-(Ω) two-wire transmission line. Design a quarter-wave two-wire air line with a 2-(cm) spacing to match the antenna to the 300-(Ω) line.

Z_{main}

$d = 2cm$

Z_L

$f = 200\times10^6$

P.8-24 $f = 2\times10^8$ (Hz),

$\lambda = 1.5\ (m). \longrightarrow \ell = \frac{\lambda}{4} = 0.375\ (m).$

Characteristic impedance of quarter-wire two-wire transmission line, $Z_0 = \sqrt{73\times300} = 148\ (\Omega)$.

For a lossless air line, from Eqs. (8-23) and (8-24),

$Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{\pi}\sqrt{\frac{\mu_0}{\epsilon_0}}\cosh^{-1}\left(\frac{D}{2a}\right).$

$148 = 120\cosh^{-1}\left(\frac{D}{2a}\right).$

Given $D = 2$ (cm) $\longrightarrow a$ (wire radius) $= 0.54$ (cm).

$c = \lambda.f$ → $\frac{3\times10^8}{200\times10^6} = \lambda = 1{,}5m$

Quarter wave old. için $\ell = \frac{\lambda}{4} = \frac{1{,}5}{4} = 0{,}375M$

Quarter wave transformer empedansı;

$Z_0 = \sqrt{Z_L.Z_{main}} = \sqrt{73\times300} \Rightarrow Z_0 = 148\ \Omega$

two wire airline

$Z_0 = 120\cosh^{-1}\left(\frac{D}{2a}\right)$

$D = 2cM$

a = tel yarıçapı

$148 = 120 . \cosh^{-1}\left(\frac{2}{2a}\right)$

cosh değeri verirse a'yı bulursun

$a = 0{,}54cM$